HPLTS-GAN: A High-Precision Remote Sensing Spatiotemporal Fusion Method Based on Low Temporal Sensitivity

Dajiang Lei, Qianwei Zhu, Yujia Li, Jiayang Tan, Shuailong Wang, Tingting Zhou, and Liping Zhang

***Abstract*— Remote sensing spatiotemporal image fusion is a promising approach to acquire remote sensing data with high spatial and temporal resolution. While most deep neural network-based models have demonstrated high accuracy, they heavily depend on temporal information from the dataset, necessitating a pair of coarse and fine resolution image sets ($C_1 - F_1$) taken near the prediction time. This reliance on temporal data makes these models challenging in practical applications. To tackle this challenge, this study introduces a time-insensitive spatiotemporal fusion model. An adaptive spatial distribution transformation (ASDT) module is proposed to enhance the spatiotemporal consistency of fine-resolution images by incorporating the spatial structure of coarse-resolution images. This module aims to improve the model's performance in tasks less sensitive to time. Furthermore, an encoding-decoding structure is devised to address significant resolution variations in remote sensing images. The encoder uses a multilevel feature extraction (MLFE) module to capture features at multiple levels, reducing information loss and enhancing feature utilization. The decoder includes a cross-scale attention feature fusion and reconstruction (CSAFFIR) module to integrate features of different scales and semantic levels, thereby improving the overall performance of the fused image. Experimental findings from datasets collected by two satellite types demonstrate that the proposed HPLTS-GAN model surpasses existing two-input models in subjective and objective evaluations. In addition, our approach shows competitiveness with current three-input models.**

***Index Terms*— Generative adversarial networks (GANs), remote sensing images, spatiotemporal fusion.**

I. Introduction

HIGH spatial and temporal resolution remote sensing data plays a significant role in atmospheric monitoring [1], vegetation cover change monitoring [2], [3], [4], [5], [6], [7], [8], ecosystem monitoring [9], and various other applications. Due to technological limitations and budget constraints, it is often challenging to achieve both high spatial and high temporal resolution imagery within a single remote sensing satellite sensor [10]. In recent years, researchers have developed a considerable number of spatiotemporal fusion methods [10], [11]. Existing methods can be categorized into weight-based approaches [12], [13], [14], [15], unmixing-based methods [16], [17], [18], Bayesian methods [19], and learning-based techniques [20], [21], [22], [23].

Methods based on weight functions assume that changes in surface reflectance can be predicted by aggregating and weighting the pixel values of spatiotemporal images. The spatial and temporal adaptive reflectance fusion model (STARFM) proposed by Gao et al. [12] is the first spatiotemporal fusion method based on weight functions. STARFM assumes that all pixels in the coarse-resolution image are pure and employs a weighted strategy to predict the target image by adding reflectance changes between two coarse images to a previous fine-resolution image. STARFM and its variations are widely used methods based on weight functions. The STAARCH model, developed by Hilker et al. [13], enhances the STARFM model using the temporal breakpoint detection technique to detect moments of land cover change from a sequence of coarse-resolution images. The model provides improved performance compared to the original STARFM. Zhu et al. [14] introduced the enhanced STARFM (ESTARFM), which incorporates reflectance conversion coefficients between coarse and fine-resolution images. This enhancement leads to improved prediction accuracy in heterogeneous regions when compared to STARFM. In addition, Cheng et al. [15] introduced a spatial and temporal nonlocal filter-based data fusion method, which utilizes nonlocal filtering and two regression coefficients to describe land surface changes, yielding more accurate and robust predictions by leveraging the redundancy in fine-resolution images.

Unmixing-based techniques operate under the assumption that pixels in the coarse-resolution image are mixtures of fine-resolution endmembers. The multisensor multiresolution image fusion based on unmixing (MMT) model developed by Zhukov et al. [16] is the pioneering spatiotemporal fusion approach to employ unmixing. It classifies coarse pixels into categories for the predicted date, obtains reflectance changes for each category, and generates the final prediction based

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on the classifications in previous fine-resolution images. Niu's spatial and temporal reflectance fusion model (STDFA) introduces temporal information from multiple fine images and performs unmixing on coarse pixels to obtain reflectance changes for each category, generating fine-resolution predictions [17]. Zhu et al. [18] proposed a flexible spatiotemporal data fusion algorithm (FSDAF), combining decomposition and weighting. This algorithm calculates spectral changes in uniform regions, predicts spatial variations using interpolation, and finally derives fine-resolution images through a weighted combination of spectral and spatial features.

Bayesian-based methods build fusion models based on Bayesian principles, such as Wei's spatiotemporal fusion of MODIS and Landsat-7 reflectance images via compressed sensing, which uses maximum a posteriori probability to describe the fusion problem [19]. It constructs a comprehensive relationship model using all involved data and obtains the fused image through classical conjugate gradient optimization.

Learning-based models use existing datasets to train models that capture relationships between images [24]. These models can be categorized into sparse representation-based and deep learning-based models. Huang's spatiotemporal reflectance fusion via sparse representation (SPSTFM) is an example of a sparse representation-based model [20]. Based on deep learning models, there are several proposed methods. Song et al. [21] proposed spatiotemporal satellite image fusion using deep convolutional neural networks (STFDCNNs), which utilizes high-pass modules for super-resolution and fusion. Li et al. [22] proposes spatiotemporal fusion of remote sensing images using a convolutional neural network (CNN) with attention and multiscale mechanisms (AMNET), which enhances fusion accuracy by incorporating AMNETs. Tan et al. [23] proposed an enhanced deep convolutional model for spatiotemporal image fusion (EDCSTFN), which utilizes CNN to extract main frame and background information from high-resolution images and high-frequency components from low-resolution images and then reconstructs the prediction results by fusing these extracted features.

In recent years, with significant advancements in computing hardware performance, deep learning models have shown impressive potential [25]. These models can automatically learn complex patterns and relationships, greatly enhancing the efficiency and accuracy of computer modeling. In the field of remote sensing, particularly in the problem of remote sensing image fusion, the introduction of deep CNNs has achieved significant success. This approach has exceled in improving model accuracy and significantly increasing the accuracy of prediction images. Considering the many advantages demonstrated by GAN methods in the field of image generation, they have also found applications in remote sensing imagery. Song et al. [26] proposed a multilayer feature fusion model based on generative adversarial networks (MLFF-GANs), to address the significant resolution disparities between fine-resolution and coarse-resolution images. Liu et al. [27] proposed a progressive spatiotemporal attention fusion model (PSTAF-GAN), based on GANs with multiband training, where a flexible multiscale feature extraction (MLFE) framework is employed to thoroughly explore multiscale information and gradually reconstruct the target image. In addition, there are other models such as STFGAN [28], CycleGANSTF [29], and GAN-STFM [30].

Although the aforementioned methods have achieved good fusion accuracy in the spatiotemporal fusion field, the practical application of many of these models is challenging because obtaining the required pairs of coarse and fine-resolution images at the same time is difficult. This limitation hinders the effective application of spatiotemporal fusion models in real-world scenarios [31], [32]. Existing models that do not rely on time information struggle to fully utilize the information from both the prediction-time coarse-resolution image and the randomly timed image to produce satisfactory results. Therefore, this article presents a high-precision spatiotemporal fusion network that is less sensitive to time, addressing the need to reduce dependency on time information while improving prediction accuracy. The main contributions of this article are as follows.

1) We design an adaptive spatial distribution transformation (ASDT) module that learns the spatial dependencies between C_t and randomly timed land satellite images (F_r). This module enhances the spatiotemporal consistency of fine-resolution images, effectively addressing the issue of missing temporal information in time-sensitive spatiotemporal fusion tasks.
2) To tackle the significant resolution differences between coarse and fine-resolution images, we introduce an encoding-decoding structure in this study. The encoder component processes input images using a multilevel feature extraction (MLFE) module to better capture multilevel feature representations, thereby reducing information loss and further improving the expressive power of the model. The decoder component incorporates a cross-scale attention feature fusion and reconstruction (CSAFFIR) module, which adaptively assigns weights to shallow and deep-level feature information, effectively integrating features at multiple scales for image reconstruction. This structure not only enhances feature learning capabilities but also demonstrates strong generalization abilities.
3) Through comparative experiments on the classic Gwydir downstream basin (LGC) dataset and the Coleambally irrigation area (CIA) dataset, our HPLTS-GAN model performs exceptionally well, outperforming other excellent spatiotemporal fusion models. This highlights the superiority of our proposed model in addressing spatiotemporal fusion tasks on real-world datasets.

The organization of the rest of this article is as follows. In Section II, we provide a brief introduction to spatiotemporal fusion models and generative adversarial networks (GANs). Section III details the proposed method. Section IV presents experimental results using simulated and real-world data, including comparisons with other advanced models. Finally, in Section V, we summarize the findings and contributions of this article.

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II. Related Work

A. Spatiotemporal Fusion Model

Spatiotemporal fusion algorithms refer to techniques that combine two or more data sources with similar spectral ranges through specific algorithms to generate richer information than that provided by individual data sources. This process involves retrieving temporal and spatial ground changes from coarse-resolution images and extracting detailed ground textures from fine-resolution images to synthesize new data with enhanced spatial and temporal resolution.

In spatiotemporal fusion tasks, it is common practice to use a pair of fine-resolution images at time t_1 (F_1) and a coarse-resolution image (C_1) and a coarse-resolution image at time t_2 (C_2) to predict the fine-resolution image at time t_2 (F_2). It is important that the time interval between t_1 and t_2 is kept as small as possible, and during this period, there should not be significant changes in the land cover or land features. Three images are input into the network to predict the fine-resolution image (F_2) at time t_2. Fig. 1(a) shows the prediction of the three-input model. The entire model prediction process can be expressed by

$$F_2 = f(C_1, F_1, C_2). \tag{1}$$

To reduce the model's dependency on time, the model's input images consist of only one fine-resolution image at any time (F_r) and a coarse-resolution image at the prediction time t (C_t). These two images are input to the model to predict the fine-resolution image at the prediction time t (F_t). Fig. 1(b) shows the prediction of the two-input model. The entire model prediction process can be expressed by

$$F_t = f(C_t, F_r). \tag{2}$$

Due to the ability of the three-input model to leverage more data and information, it has an advantage over the two-input model, resulting in higher accuracy in predicting images. Therefore, in the two-input model, it is necessary to fully utilize the image information to compensate for the lack of data caused by the absence of one input image.

B. Generative Adversarial Network

GAN is inspired by game theory and comprises a generator and a discriminator, creating a competitive relationship between them [33]. Its core idea revolves around the continuous improvement of performance through the adversarial process between the generator and the discriminator, making the generated data more and more realistic. GAN has made significant breakthroughs in the field of computer vision, including image generation [34], image super-resolution [35], style transfer [36], and more. Specifically, the generator (G) takes a random noise vector z and attempts to generate data x, while the discriminator (D) evaluates x for dissimilarity to real data and outputs a matrix in the range of 0–1 to judge the authenticity of the data. GAN engages in a continual adversarial training process until the generated data is convincing enough that the discriminator cannot effectively distinguish between real and generated data, thus making the generated data distribution as close as possible to p_{data}.

Due to the GAN network's generator taking a random noise vector as input and attempting to generate data samples similar to real data, which the discriminator distinguishes between real and generated data. These two components compete with each other, enabling the generator to enhance its ability to generate fine-resolution images as it approaches the final prediction stage. This enables the reduction of temporal dependency on reference images while still achieving good fusion accuracy [30]. In the training of the HPLTS-GAN model, we choose the least squares GAN (LSGAN) as the adversarial loss function for generation [33], as follows:

$$\min_G \max_D \mathcal{L}_{\text{LSGAN}}(D, G) = E_{x\sim p_{\text{data}}(x)}\left[\log(G(z))\right] + E_{z\sim p_z(z)} \times \left[\log(1 - D(G(z)))\right] \tag{3}$$

$$\min_D \mathcal{L}_{\text{LSGAN}}(D) = \frac{1}{2} E_{x\sim p_{\text{data}}(x)}\left[(D(x) - b)^2\right]$$

$$\min_G \mathcal{L}_{\text{LSGAN}}(G) = \frac{1}{2} E_{z\sim p_z(z)}\left[\left(D(G(z) - c)^2\right)\right] \tag{4}$$

where a, b, and c are constants. When $b-c = 1$ and $b-a = 2$, the object function is equivalent to Pearson χ^2 divergence.

III. Method

The task of this article is to predict using two input images, specifically, utilizing a randomly timed fine-resolution image (F_r) and a coarse-resolution image at time t (C_t) to predict the fine-resolution image (F_t) at time t. In Tan et al.'s approach, the features extracted from F_r and C_t are simply combined without fully exploiting the features from both images. Therefore, this article introduces a GAN-based model named HPLTS-GAN. The overall framework of HPLTS-GAN is illustrated in Fig. 2, and the model comprises two main components: the generator and the discriminator. The input to the generator includes C_t and F_r, aiming to comprehensively fuse and reconstruct the predicted image $\hat{F}_t$ by leveraging the global features from C_t and the detailed features from F_r. The predicted $\hat{F}_t$ and the real image F_t are then combined separately with C_t to form two sets of inputs for the discriminator. The discriminator's role is to discern between the real and generated images, pushing the generated image $\hat{F}_t$ closer to the real image F_t. This adversarial training process iteratively refines the generator's ability to produce more realistic predictions. In the following, the proposed HPLTS-GAN will be addressed in detail.

A. Generator

To address the issue of missing temporal information in image inputs and enhance the utilization of input information, the generator part of HPLTS-GAN comprises three key modules, as illustrated in Fig. 3. The left module is the ASDT module, which aims to improve the spatiotemporal consistency of fine-resolution images by synthesizing the spatial structure of coarse-resolution images. The middle module is the MLFE module, and the right module is the CSAFFIR module, together forming an encoder–decoder structure. To comprehensively and finely extract features from images, we utilize the MLFE module to capture detailed representations of multilevel features in remote sensing images. For the purpose of

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Fig. 1. Concept of spatiotemporal fusion model prediction. (a) Prediction concept for the three-input model, where the input consists of the coarse-resolution image at the prediction time and a pair of coarse-fine resolution images taken at a similar time. (b) Prediction concept for the two-input model, where the input consists of the coarse-resolution image at the prediction time and a fine-resolution image captured at random time.

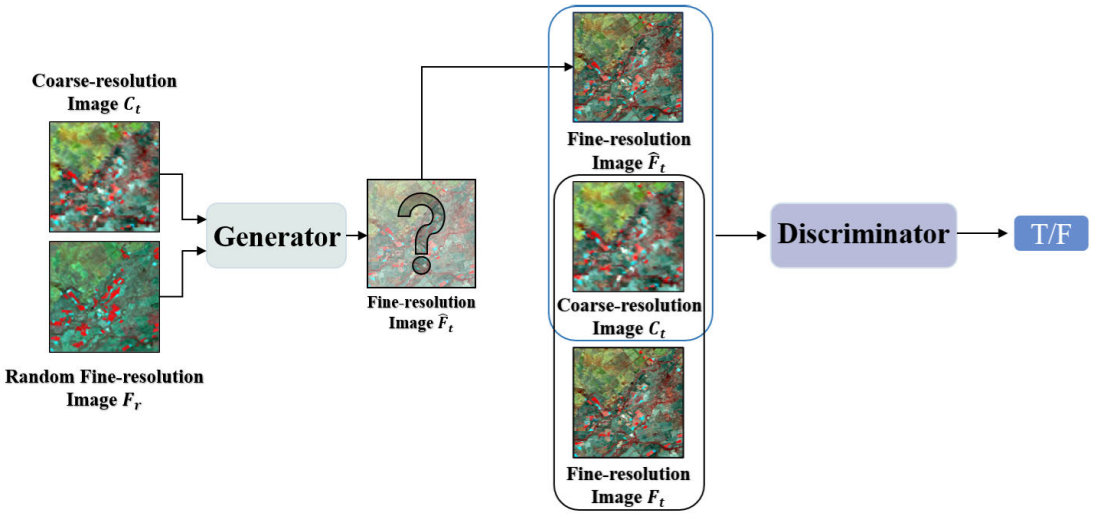


Fig. 2. Overall framework of HPLTS-GAN. The generator is responsible for image generation. The discriminator's role is to distinguish between real and fake images.

fusing feature maps of different scales, each having distinct semantic levels, and subsequently reconstructing the image, the CSAFFIR module is designed to adaptively assign weights to shallow and deep-level feature information. This design enhances the fusion image's quality.

Due to the different time periods captured by F_r and C_t, changes in land features may occur due to natural disasters such as earthquakes or floods. As a result, the spatial distribution of objects in the two images may be inconsistent. Therefore, simply integrating the features of these two images makes it challenging to accurately capture the global feature distribution of F_t, leading to a decrease in prediction accuracy.

To address this issue, we design the ASDT module in this study. ASDT learns the cross-feature spatial dependencies between C_t and F_r and transforms F_r to become a fine-resolution image with the spatial feature distribution of C_t. This process aims to enhance the spatiotemporal consistency of fine-resolution images by synthesizing the spatial structure of coarse-resolution images, effectively mitigating the problem mentioned. The ASDT module as shown in Fig. 4. First, both F_r and C_t are input into the ASDT module. The input images (Input ϵ $\mathbf{R}^{C\times H\times W}$) are transformed into feature matrices (M ϵ $\mathbf{R}^{C\times H}$) through convolution. The feature map (M_Q) is obtained from C_t, while the feature matrices (M_K) and (M_V) are obtained from F_r. M_Q serves as the query, and M_K as the key to model the spatial relationships between any two points in M_Q and M_K, learning the spatial dependencies between M_Q and M_K. This is achieved through a softmax layer that normalizes the numerical vectors to obtain weight vectors, resulting in the desired spatial attention map ($\text{Feature}_{\text{map}}$). The $\text{Feature}_{\text{map}}$ is used for matrix operations with M_V, and the resulting matrix is mapped back to the size of the input feature map. A residual operation is performed with the original input F_r, resulting in F_r^t. This process is described as follows:

$$M = L(\sigma(\text{conv}(\text{Input}))) \tag{5}$$

$$\text{Feature}_{\text{map}} = \delta\left(M_Q \otimes M_K\right) \tag{6}$$

$$F_r^t = L\left(\text{Feature}_{\text{map}} \otimes M_V\right) \oplus F_r \tag{7}$$

where M denotes the feature matrix, $\text{conv}(\cdot)$ denotes a 7×7 convolution operation, $\sigma(\cdot)$ denotes a 1×1 convolution operation, Input denotes the input image, $L(\cdot)$ denotes a linear transformation operation, $\delta(\cdot)$ denotes the softmax operation, $\text{Feature}_{\text{map}}$ denotes the spatial attention map, $\otimes$ denotes element-wise multiplication, and $\oplus$ denotes element-wise addition.

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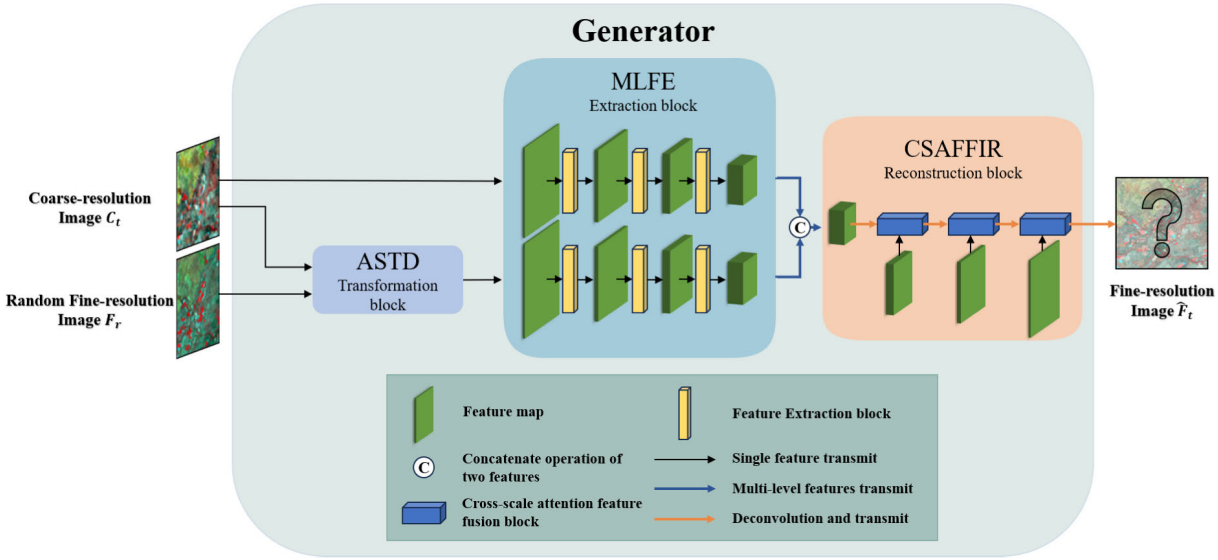


Fig. 3. Overall framework diagram of the generator. The ASTD module on the left is employed for the spatial feature transformation of randomly fine-resolution images, the MLFE module in the middle is used for extracting comprehensive feature information from the image, and the CSAFFIR module on the right is applied for feature fusion and image reconstruction.

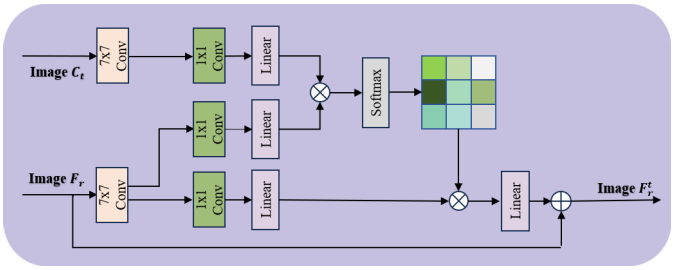


Fig. 4. ASDT module.

In this article, the encoder–decoder structure designed is similar to the architecture of u-net [37]. In order to prevent gradient dispersion and gradient explosion to ensure the training of the model, we introduce a basic block similar to the residual block in the residual network (ResNet) [38] in both the encoder and decoder parts. This basic block eliminates batch normalization to improve the quality of model-generated images [39].

In the encoder part, the MLFE module is used to extract coarse-resolution and fine-resolution image features in detail. The whole feature extraction process is divided into four stages. In order to make the image have complete features to ensure the accuracy of the model, the downsampling operation is not performed in the first stage. In the remaining stages, 3×3 convolution kernel is used for downsampling to enlarge the receptive field, obtain deeper information, and fully extract the image spectral and structural features of the images. The feature extraction module containing downsampling operations is shown in Fig. 5. The entire feature extraction process can be expressed as follows:

$$F_x^i = D\left(F_x^{i-1}\right) + R\left(D\left(F_x^{i-1}\right)\right) \tag{8}$$

where $D(\cdot)$ represents the downsampling operation, $R(\cdot)$ represents the residual basic block, and F_x^i represents the features extracted in the ith stage of image x, where $i = 0, 1, 2, 3$.

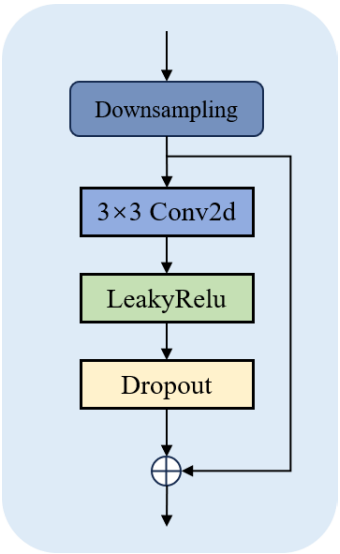


Fig. 5. Feature extraction module. The three middle blocks from the basic building blocks of the residual block.

Therefore, the coarse-resolution image C_t, after feature extraction, obtains a feature sequence $[F_{C_t}^0, \ldots, F_{C_t}^3]$, and the fine-resolution image F_r, after feature extraction, obtains a feature sequence $[F_{F_r}^0, \ldots, F_{F_r}^3]$. These two sequences are concatenated level by level to obtain a multilevel coarse-fine resolution feature sequence $[F_{C-F}^0, \ldots, F_{C-F}^3]$.

The decoder part of CSAFFIR is responsible for fusing feature maps of different scales and semantic levels and performing image reconstruction. The attention feature fusion component in CSAFFIR incorporates the multiscale channel attention module (MS-CAM) [40], and the specific process is illustrated in Fig. 6. The two inputs come from feature maps of different scales. MS-CAM has two branches to aggregate global and local contextual information. One branch uses global average pooling to extract attention for global features,

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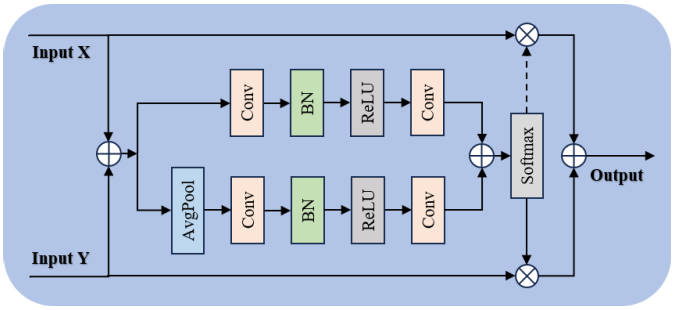


Fig. 6. Attention feature fusion module. The middle two branches form the MS-CAM.

while the other branch directly employs convolution to extract channel attention for local features. By attending to both global and local features, weights are computed and assigned to the input feature maps. This feature fusion process can be represented as follows:

$$F^i = A\left(F^i_{C-F} \oplus D\left(F^{i-1}_{C-F}\right)\right) \otimes F^i_{C-F} \oplus \left(1 - A\left(F^i_{C-F} \oplus D\left(F^{i-1}_{C-F}\right)\right)\right) \otimes F^{i-1}_{C-F} \quad (9)$$

where F^i denotes the output features after feature fusion, $A(\cdot)$ denotes the MS-CAM module, $D(\cdot)$ denotes the deconvolution operation, $\oplus$ denotes feature integration, and F^i_{C-F} denotes the result obtained by concatenating the features extracted from the ith stage of coarse-fine resolution images, where $i = 0, 1, 2, 3$.

The image reconstruction stage and the feature extraction stage are symmetrical and are also divided into four stages. Except for the first stage, all subsequent stages involve cross-scale feature fusion. This is because features at different levels contribute differently to the final result. Simply adjusting features from different layers to the same size for concatenation may not achieve higher-precision image reconstruction. Furthermore, since the last three stages of feature extraction involve downsampling to extract high-level semantic information, there is a loss of some information. Therefore, performing cross-scale feature fusion during the image reconstruction process selectively focuses on features with different scales and semantics, facilitating their fusion. This allows the model to preserve high-level semantic information while better-identifying details and improving overall performance.

B. Discriminator

As shown in Fig. 7, the discriminator in HPLTS-GAN is a simple binary CNN. Each block of the discriminator consists of multiple convolutional layers, instance normalization layers [41], and leaky rectified linear unit (LeakyReLU) activation functions [42]. During the training of the discriminator, its input is the coarse-resolution image C_t at the predicted time t and the real fine-resolution image F_t (or the model's prediction of the fine-resolution image $\hat{F}_t$ at time t). When the input is C_t and F_t, the discriminator expects to get an all-real matrix, and when the input is C_t and $\hat{F}_t$, it expects to get an all-fake matrix. The discriminator has significantly fewer parameters than the generator because it only needs to abstract and generalize through a deep network. The adversarial function provided by the discriminator helps the generator obtain higher quality results.

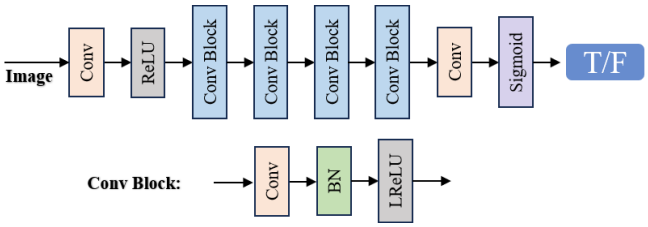


Fig. 7. Overall framework diagram of the discriminator.

C. Loss Function

The generator's loss consists of four components: GAN loss ($\mathcal{L}_{\text{LSGAN}}$), L1 loss ($\mathcal{L}_1$), spectral angle loss ($\mathcal{L}_{\text{Spectrum}}$), and vision loss ($\mathcal{L}_{\text{Vision}}$). The total loss can be denoted as

$$\mathcal{L}(G) = \alpha * \mathcal{L}_{\text{LSGAN}} + \beta * \mathcal{L}_1 + \eta * \mathcal{L}_{\text{Spectrum}} + \theta * \mathcal{L}_{\text{Vision}} \quad (10)$$

where α, β, η, and θ are weighting coefficients.

$\mathcal{L}_{\text{LSGAN}}(G)$ is the generator's loss function. The other three losses, $\mathcal{L}_1$ loss, $\mathcal{L}_{\text{Spectrum}}$, and $\mathcal{L}_{\text{Vision}}$ are image-related losses that ensure the authenticity of the generated images. $\mathcal{L}_1$ is used to ensure that the values of the generated image match those of the input image at the same pixel positions, as shown in (10). $\mathcal{L}_{\text{Spectrum}}$ represents the cosine similarity between the real image and the generated image, aiming to capture the spectral fidelity of the generated remote sensing image, reducing and controlling spectral distortion between different spectral bands, as shown in (11). $\mathcal{L}_{\text{Vision}}$ is computed using the multiscale structural similarity (MS-SSIM) [43] to ensure the structural fidelity of the generated remote sensing image, as shown in (12)

$$\mathcal{L}_1 = \frac{1}{K}\sum_{k=1}^{K} \|\mathbf{T} - \mathbf{P}\|_1 \quad (11)$$

$$\mathcal{L}_{\text{Spectrum}} = \mathbf{I} - \frac{\mathbf{T}\cdot\mathbf{P}}{\|\mathbf{T}\|_2 \|\mathbf{P}\|_2} \quad (12)$$

$$\mathcal{L}_{\text{Vision}} = \mathbf{I} - [\mathbf{l_m}(\mathbf{T},\mathbf{P})]^{\nu_N} \cdot \prod_{\mathbf{j}=\mathbf{1}}^{\mathbf{N}} \left[\mathbf{c_j}(\mathbf{T},\mathbf{P})\right]^{\rho_j} \left[\mathbf{S_j}(\mathbf{T},\mathbf{P})\right]^{\mu_j} \quad (13)$$

where T represents the real image matrix (e.g., F_t), P represents the predicted image matrix (e.g., $\hat{F}_t$), $\|\cdot\|_P$ is the P-norm, K is the number of image pixels, and I is a tensor of ones, N is the index of the highest scale, and l_j, c_j, and s_j represent the luminance, contrast comparison, and structure comparison for the jth scale, respectively. v_N, ρ_j, and μ_j are the corresponding weight coefficients.

IV. Experiment

A. Study Areas and Datasets

To evaluate the effectiveness of our proposed method, we conduct experiments on two publicly available datasets.

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The first dataset is located in the CIA in southern New South Wales, Australia (34.0034°E, 145.0675°S). This area is an irrigation system for various summer crops, with phenological changes being dominant. From October 2001 to May 2002, a total of 17 cloud-free Landsat-MODIS image pairs were collected. Each image consists of six spectral bands, with an image size of 3200 × 2720, using Landsat 7 ETM+ and MODIS Terra MOD09GA 6 reflectance data. The second dataset is from the Lower Gwydir Catchment (LGC) in Australia (149.2815°E, 29.0855°S). From April 2004 to April 2005, a total of 14 cloud-free image pairs were obtained, containing Landsat 5 TM and MODIS Terra MOD09GA 6 reflectance data. The image size is 1720 × 2040. Due to flood events, the LGC dataset can be considered as a time-dynamic landscape. The datasets have undergone atmospheric correction, and MODIS images are upsampled to a spatial resolution of 25 m using bilinear interpolation. Subsequently, each Landsat-MODIS pair is cross-verified within a 25 m pixel range. We crop the datasets to a size of 1200 × 1200, ensuring consistency among the Landsat-MODIS image pairs under study.

B. Experimental Setting

The proposed method is compared with several commonly used spatiotemporal fusion techniques, including the weighted function algorithm STARFM, the hybrid-based FSDAF, and deep learning-based spatiotemporal fusion models EDCSTFN, GAN-STFM, MLFFGAN, and AMNET. Except for the GAN-STFM model, all other models require at least one pair of coarse and fine-resolution images and the coarse-resolution image of the predicted date as input for prediction. In contrast, the HPLTS-GAN and GAN-STFM models only require a random-time fine-resolution image and the coarse-resolution image of the predicted date as input. The training data for STARFM, FSDAF, EDCSTFN, MLFFGAN, and AMNET models are grouped into pairs of Landsat-MODIS data with acquisition dates that are sufficiently close. Each group of data consists of two Landsat-MODIS data pairs. The training data for HPLTS-GAN and GAN-STFM models do not require grouping. The input images for these models only consist of the coarse-resolution image at the prediction time and a randomly selected fine-resolution image from the Landsat dataset in the study area.

For quantitative evaluation, we use five metrics to assess the image fusion results. The spectral angle mapper (SAM) is used to measure the spectral distortion in the images. A smaller SAM value indicates a better preservation of spectral information in the generated image. The structural similarity index (SSIM) is used to measure the overall structural similarity between the generated and real images, with a higher SSIM value indicating better fusion results. The root mean square error (RMSE) assesses the overall difference between the original and fused images, where a lower RMSE value signifies better fusion accuracy. The relative dimensionless global error (ERGAS) assesses the spectral quality of the fused image in all spectral bands, with a lower ERGAS value signifying better spectral quality. The correlation coefficient (CC) measures the linear correlation between the original and fused images, with a higher CC value indicating a stronger relationship between the two images. This method can be implemented using the Python-based PyTorch framework.

TABLE I
RESULTS OF CONTRAST EXPERIMENT ON THE LGC DATASET

Date	Model	SAM	SSIM	RMSE	ERGAS	CC
Feb.14th	STARFM	3.3686	0.7530	0.0473	0.6967	0.9326
	FSDAF	3.0532	0.8434	0.0463	0.7036	0.9310
	EDCSTFN	4.0106	0.7540	0.0381	0.8533	0.8837
	GAN-STFM	3.4566	0.8010	0.0452	0.6977	0.9172
	MLFF-GAN	4.8026	0.6222	0.0607	1.0337	0.8298
	AMNET	3.5828	0.7330	0.0438	0.7695	0.9067
	Ours	**2.9522**	**0.8600**	**0.0338**	**0.6414**	**0.9326**
Mar.2nd	STARFM	3.0558	0.7893	0.0401	0.6499	0.9372
	FSDAF	2.7826	0.8730	0.0364	0.6385	0.9285
	EDCSTFN	4.1679	0.7781	0.0387	0.8737	0.8820
	GAN-STFM	3.4210	0.8151	0.0383	0.6920	0.9120
	MLFF-GAN	4.4773	0.6752	0.0521	1.0030	0.8315
	AMNET	3.4037	0.8307	0.0346	0.7718	0.8980
	Ours	**2.7584**	**0.8875**	**0.0263**	**0.5848**	**0.9392**
Apr.3rd	STARFM	3.8605	0.7084	0.0352	0.8743	0.8258
	FSDAF	3.8031	0.7907	0.0340	0.8633	0.8330
	EDCSTFN	3.2172	0.7592	0.0310	0.8597	0.8283
	GAN-STFM	3.7300	0.7524	0.0325	0.8421	0.8289
	MLFF-GAN	3.7844	0.7078	0.0323	0.9232	0.8135
	AMNET	3.3370	0.7695	0.0344	0.9338	0.8115
	Ours	**3.0398**	**0.8235**	**0.0366**	**0.6985**	**0.8705**
Mean	STARFM	3.4283	0.7502	0.0409	0.7403	0.8985
	FSDAF	3.2130	0.8357	0.0389	0.7351	0.8975
	EDCSTFN	3.7986	0.7638	0.0359	0.8622	0.8647
	GAN-STFM	3.5359	0.7895	0.0386	0.7439	0.8860
	MLFF-GAN	4.3548	0.6684	0.0483	0.9866	0.8250
	AMNET	3.4412	0.7777	0.0376	0.8250	0.8721
	Ours	**2.9168**	**0.8570**	**0.0322**	**0.6416**	**0.9141**
Reference		0	1	0	0	1

We crop the MODIS and Landsat images to a uniform size of 128 × 128 as model inputs and set the sliding stride to 64. The initial learning rate (lr) of the EDCSTFN model is 1e-03, while the initial lr of other deep learning models is 2e-04. The batch size for deep learning models is set to 32, and the number of iterations is 200. Our experiments are conducted on a computer with an Intel Xeon Gold 6148 CPU 2.40 GHz and an NVIDIA Tesla V100-SXM2-16 GB graphics processor.

C. Experimental Results

1) Metric Evaluation: The results of experiments for the three datasets in the LGC region in 2005 and three datasets in the CIA region in 2002, using various algorithms for image fusion, are obtained to evaluate the accuracy of the models. The quantitative evaluations of the different spatiotemporal fusion methods for the LGC dataset are shown in Table I and for the CIA dataset in Table II.

For the LGC dataset as shown in Table I, in most cases, our proposed HPLTS-GAN displays better performance. In traditional methods, the overall performance of the FSDAF method is superior to the STARFM method, possibly because the FSDAF method has a better ability to adapt to heterogeneous

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TABLE II
RESULTS OF CONTRAST EXPERIMENT ON THE CIA DATASET

Date	Model	SAM	SSIM	RMSE	ERGAS	CC
Apr.18th	STARFM	5.1749	0.6845	0.0571	1.2151	0.7975
	FSDAF	5.5088	0.7093	0.0519	1.2978	0.7893
	EDCSTFN	4.5988	0.7138	0.0369	1.1550	0.8200
	GAN-STFM	4.3285	0.7163	0.0367	1.1115	0.8203
	MLFF-GAN	4.1858	0.7013	0.0403	1.1336	0.8102
	AMNET	4.4717	0.7090	0.0425	1.1494	0.8076
	Ours	**3.9151**	**0.8038**	**0.0334**	**1.0180**	**0.8572**
Apr.27th	STARFM	4.5733	0.6854	**0.0507**	1.2720	0.7883
	FSDAF	4.6811	0.7074	0.0548	1.3281	0.7817
	EDCSTFN	3.7295	0.7387	0.0893	1.0857	0.8261
	GAN-STFM	3.9116	0.7537	0.0731	1.0308	0.8282
	MLFF-GAN	3.7721	0.7398	0.0936	1.0964	0.8136
	AMNET	3.6660	0.7382	0.1023	1.0790	0.8127
	Ours	**3.6696**	**0.8193**	0.0595	**1.0475**	**0.8680**
May.4th	STARFM	3.9295	0.7746	0.0427	0.9341	0.8536
	FSDAF	4.0788	0.7949	0.0427	1.0051	0.8465
	EDCSTFN	**2.9966**	0.8382	**0.0261**	**0.8337**	0.8819
	GAN-STFM	3.3849	0.8177	0.0284	0.8719	0.8673
	MLFF-GAN	3.1577	0.8170	0.0289	0.8740	0.8733
	AMNET	3.0836	0.8283	0.0260	0.8764	0.8740
	Ours	3.2238	**0.8826**	0.0356	0.8539	**0.9023**
Mean	STARFM	4.5592	0.7148	0.0502	1.1404	0.8132
	FSDAF	4.7562	0.7372	0.0498	1.2103	0.8059
	EDCSTFN	3.7750	0.7636	0.0508	1.0248	0.8426
	GAN-STFM	3.8750	0.7625	0.0461	1.0047	0.8386
	MLFF-GAN	3.7052	0.7527	0.0543	1.0347	0.8323
	AMNET	3.7404	0.7585	0.0570	1.0349	0.8315
	Ours	**3.6028**	**0.8352**	**0.0428**	**0.9731**	**0.8758**
Reference		0	1	0	0	1

region changes compared to the STARFM method. In some cases, the quantitative metrics of the FSDAF method exceed those of some deep learning methods. We believe this might be due to the obvious features present in the LGC dataset, making pixels easy to classify into the correct categories. For the FSDAF method, these categories are the final members in its spectral decomposition, thus making the LGC dataset more favorable for FSDAF. Among the deep learning-based models, EDCSTFN, MLFF-GAN, and AMNET are three-input models, while GAN-STFM and the proposed HPLTS-GAN in this article are two-input models. In three-input models, AMNET overall outperforms EDCSTFN and MLFF-GAN, with the superiority being most evident in SAM and ERGAS metrics. When comparing EDCSTFN and MLFF-GAN, EDCSTFN's overall performance is better than MLFF-GAN.In two-input models, HPLTS-GAN performs better than GAN-STFM, attributed to HPLTS-GAN effectively utilizing the global similarity information between the two input images and extracting and fusing features more finely, thereby compensating for the information loss caused by fewer inputs. On the LGC dataset, the two-input models HPLTS-GAN and GAN-STFM outperform the three-input models.

For the CIA dataset as shown in Table II, the overall performance of the traditional method FSDAF is superior to the STARFM method. Models based on deep learning outperform traditional methods overall. Among the three-input models, EDCSTFN, MLFF-GAN, and AMNET exhibit different advantages in different data groups. In the case of two-input models, HPLTS-GAN still outperforms GAN-STFM overall. Among deep learning models, HPLTS-GAN shows better overall fusion metrics performance, while EDCSTFN and GAN-STFM models exhibit superior performance in certain metrics for certain data.

In general, most models perform better on the LGC dataset compared to the CIA dataset, possibly because the images in the CIA are more complex. Overall, deep learning-based models outperform traditional methods. Among the three-input models, EDCSTFN and AMNET perform well on both datasets, but their advantage over two-input models is smaller on the LGC dataset. This might be because the LGC dataset exhibits significant changes in surface details, making it more challenging for EDCSTFN to handle compared to two-input models. MLFF-GAN performs notably better on the CIA dataset, likely because the multilayer structure of U-Net in MLFF-GAN is more effective on the complex images in the CIA dataset but struggles to accurately predict abrupt changes in the LGC dataset. Among the two-input models, GAN-STFM performs better on the LGC dataset, possibly because it excels at handling images with sudden changes in surface details. However, due to insufficient utilization of image information, GAN-STFM struggles to accurately fuse complex image data in the CIA dataset. HPLTS-GAN, based on time-insensitive features, maintains superior prediction accuracy on both datasets, effectively addressing the fusion challenges posed by abrupt and complex image data, thus exhibiting better robustness. Compared to the two-input model GAN-STFM, HPLTS-GAN shows significant improvement in both datasets. Moreover, compared to the time-dependent three-input models EDCSTFN, MLFF-GAN, and AMNET, HPLTS-GAN demonstrates strong competitiveness, balancing the need to reduce dependence on time information and maintain prediction accuracy.

2) Visual Comparison: To better intuitively compare the visual effects of different algorithms, Fig. 8(a) displays fusion images generated by various methods for the LGC dataset on February 14th. These images are of size 1200 × 1200, providing an overview of the global effects of different algorithms. Fig. 8(b) shows a zoomed-in view area (highlighted by yellow boxes) of the images from Fig. 8(a), focusing on the local details to observe differences in algorithm performance. STARFM and FSDAF algorithms exhibit rich color representation, and the overall distribution of the fusion images is good. However, these images suffer from texture blurring and missing details. In the three-input models, EDCSTFN and AMNET exhibit similar visual performance, with good spectral information but less ideal performance in detail texture, often appearing blurry. MLFF-GAN's visual results are unsatisfactory in both spectral and detail texture information, with blurry textures and missing details at the edges. Among the two-input models, both GAN-STFM and HPLTS-GAN generate clear images with good restoration quality. However, HPLTS-GAN excels in presenting texture and detail information, demonstrating superior visual effects.

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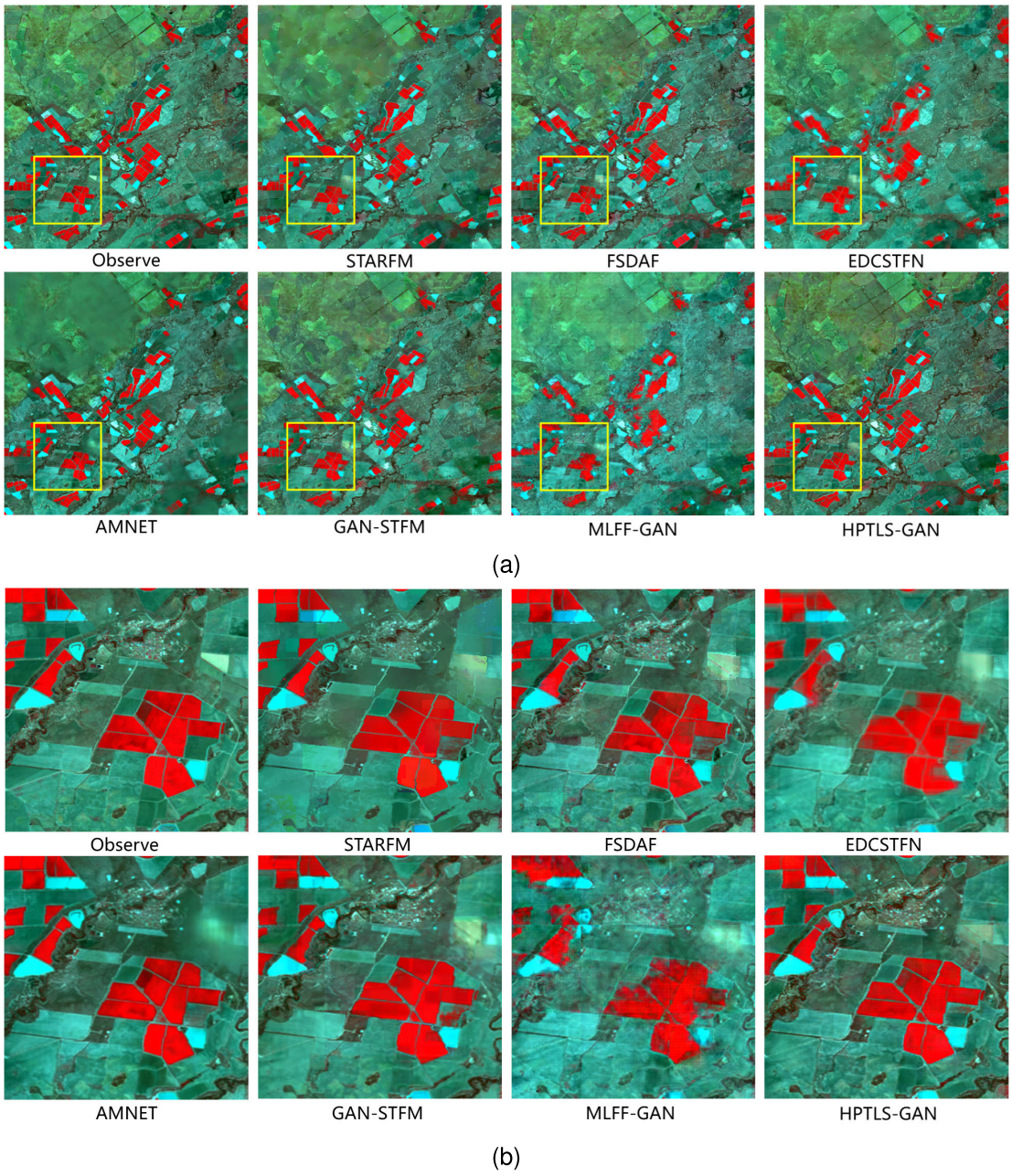


Fig. 8. Visual comparison of LGC test data on February 14th. (a) Global visual comparison. (b) Local visual comparison.

Fig. 9(a) displays fusion images generated by different methods for the CIA dataset on April 18th. These images are of size 1200 × 1200, providing an overview of the global effects of different algorithms. Fig. 9(b) shows a zoomed-in view area (highlighted by yellow boxes) of the images from Fig. 9(a), focusing on the local details to observe differences in algorithm performance. STARFM and FSDAF algorithms exhibit rich color representation, but the generated images suffer from texture blurring, missing details, and overall rough fusion effects. In the three-input models, the EDCSTFN model's generated images can effectively restore prominent features, but there are certain blurriness and spectral inaccuracies in texture details, leading to noticeable differences from the real observed images. MLFF-GAN and AMNET models produce images with less disparity from real images, with spectral information more closely resembling real observed images. However, they suffer from varying degrees of texture blurriness and insufficiently clear detail edges. Among the two-input models, GAN-STFM's generated images on the CIA dataset are less ideal, showing poor performance in both spectral information and texture details. On the other hand, HPLTS-GAN effectively presents texture and detail information, exhibiting excellent visual effects.

Overall, the visual effects of each algorithm align well with the previously discussed evaluation metrics. The proposed HPLTS-GAN method not only performs exceptionally well in terms of spectral information but also vividly presents detailed

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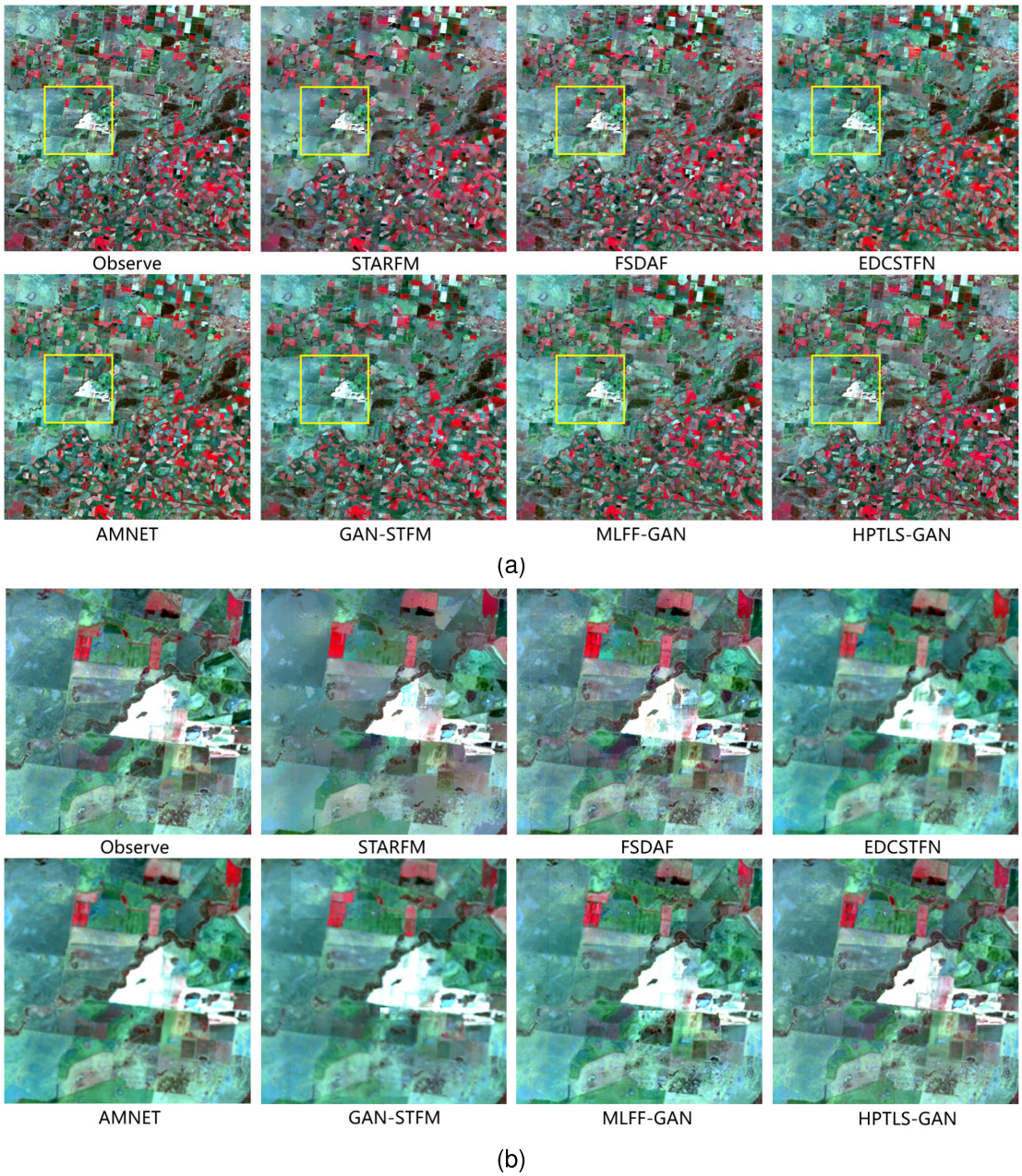


Fig. 9. Visual comparison of CIA test data on April 18th. (a) Global visual comparison. (b) Local visual comparison.

texture information, making the overall fusion result closer to real images.

3) Residual Map Comparison: Residual maps can further validate the effectiveness of the proposed method. Residual maps are typically obtained by pixel-wise subtraction of the real image from the fusion images generated by different comparison methods. The average of the residual results across all bands is then calculated to create the residual map. The fewer red contents in the residual map, and the lighter the red blocks' color, the smoother the overall map, indicating that the fusion image generated by the remote sensing spatiotemporal fusion method is more similar to the real image in terms of spectral information. In addition to spectral information, the residual map can also be used to observe whether the texture details of the image are clear. The residual comparison maps for the LGC dataset on February 14th and the CIA dataset on April 18th are shown in Figs. 10 and 11, respectively.

Observing Fig. 10, traditional methods like STARFM and FSDAF exhibit many bright red blocks, indicating more noticeable spectral errors. GAN-STFM and MLFF-GAN show more light red blocks, representing smaller errors. EDCSTFN, AMNET, and HPLTS-GAN produce smoother images with fewer red blocks. Upon close observation, it can be noticed that the images generated by EDCSTFN and AMNET are slightly blurry, indicating the presence of missing detail information. HPLTS-GAN's residual map is smoother with a better representation of texture details, and some regions have no red blocks, indicating minimal spectral errors in those areas, making them very similar to the real image's spectral information.

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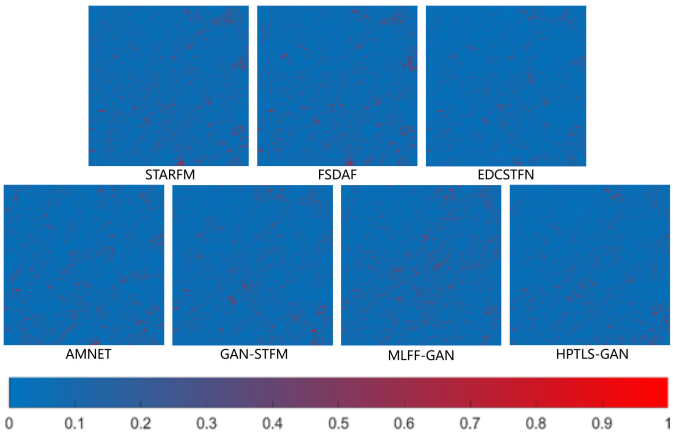


Fig. 10. Residual map comparison of LGC test data on February 14th.

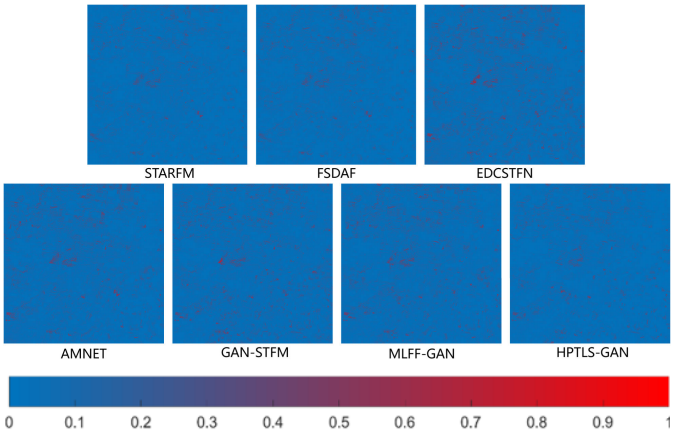


Fig. 11. Residual map comparison of CIA test data on April 18th.

Observing Fig. 11, the smoothness of the residual maps among different methods is relatively consistent, with varying degrees of red blocks in the central part of the residual maps. STARFM and FSDAF exhibit consistent performance with noticeable red blocks, and the overall residual maps are blurred with missing texture details. The residual maps of EDCSTFN, GAN-STFM, MLFF-GAN, AMNET, and HPLTS-GAN exhibit significantly better texture detail compared to traditional models. The residual maps of EDCSTFN and GAN-STFM display slightly more bright red patches, indicating larger spectral errors in localized regions for these two models. MLFF-GAN and AMNET show similar performance, appearing smoother compared to EDCSTFN and GAN-STFM, but still have more pale red patches. Therefore, MLFF-GAN and AMNET models exhibit better overall spectral information fitting and smaller spectral errors compared to EDCSTFN and GAN-STFM but still exhibit localized spectral errors. HPLTS-GAN, on the other hand, appears smoother overall with fewer red patches, indicating a closer resemblance between the fused image and the real image in terms of spectral information. In addition, richer texture details can be observed in the residual maps of HPLTS-GAN. Consequently, HPLTS-GAN demonstrates good fusion performance in both spectral information and texture detail information.

Comparing the residual maps of various models, we observe that although STARFM and FSDAF models exhibit good spectral information, they suffer from blurry texture. Among the three-input models, EDCSTFN performs well on the LGC dataset but shows spectral deviations on the CIA dataset, indicating slightly lower generalizability. MLFF-GAN exhibits subpar performance on the LGC dataset, displaying several spectral errors, whereas it demonstrates improved results on the CIA dataset. However, it still faces issues with model generalizability. AMNET performs equally well on both the LGC and CIA datasets, demonstrating good generalizability, although some regions exhibit spectral errors. Among the two-input models, GAN-STFM shows spectral errors on both datasets, indicating suboptimal performance. The proposed HPLTS-GAN model exhibits outstanding performance on both datasets, exceling not only in spectral information but also in texture detail, showcasing its robustness and generalizability.

4) Ablation Experiment Analysis: Conducting ablation experiments on the LGC dataset can effectively validate the effectiveness of different modules in the model. The model mainly consists of three modules: 1) multilevel feature; 2) cross-scale attention feature fusion module (CSAFF); and 3) ASDT module. The experimental results are presented in Table III.

a) Multilevel feature: In Table III, we observe a significant improvement in SAM, SSIM, ERGAS, and CC metrics when using the multilevel feature module compared to the baseline model. This indicates that downsampling operations during feature extraction lead to the loss of spectral and structural information, resulting in a decrease in these metrics. However, the use of the multilevel feature module allows effective utilization of features extracted at different stages and incorporates them into image reconstruction, thereby alleviating the issue of information loss in single-feature extraction.

b) CSAFF module: In Table III, we can observe that adding both the multilevel feature module and the CSAFF module to the model results in varying degrees of improvement compared to only adding the multilevel feature module. The SSIM and RMSE metrics show a more significant increase, indicating an enhancement in detail information. This suggests that using a simple duplication operation in the multilevel feature module, where the weights assigned to two feature maps during image reconstruction are the same (one feature map is obtained by duplication, and the other is obtained through deconvolution), may not be optimal. This is because the multilevel feature module employs a simple copy operation, leading to equal weights assigned to the two feature maps during the image reconstruction process, where one feature map is obtained through copying and the other through deconvolution. However, features extracted at different levels contribute differently to image reconstruction. Therefore, enabling the model to adaptively allocate weights and improve the utilization of feature information is essential.

c) ASDT module: In Table III, we can observe that adding the ASDT module to the model further improves all the metrics, with SAM and ERGAS showing the most significant improvements. This indicates that the ASDT module effectively enhances spectral information by learning the spatial

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TABLE III
ABLATION EXPERIMENT ON THE LGC DATASET

Date	Model			SAM	SSIM	RMSE	ERGAS	CC
	Multi-level Feature	AFF	ASDT					
Feb.14th	×	×	×	3.4262	0.8104	0.0591	0.7357	0.9058
	✓	×	×	3.2157	0.8340	0.0556	0.6892	0.9230
	✓	✓	×	3.4275	0.8497	0.0530	0.7506	0.9251
	✓	✓	✓	**2.9522**	**0.8600**	**0.0338**	**0.6414**	**0.9326**
Mar.2th	×	×	×	3.3089	0.7070	0.0461	0.7181	0.9045
	✓	×	×	2.8948	0.8523	0.0480	0.6591	0.9238
	✓	✓	×	2.8691	0.8803	0.0308	0.6644	0.9365
	✓	✓	✓	**2.7584**	**0.8875**	**0.0263**	**0.5848**	**0.9392**
Apr.3rd	×	×	×	3.4546	0.7678	0.0352	0.7879	0.8345
	✓	×	×	3.2143	0.7959	0.0361	0.7529	0.8509
	✓	✓	×	3.1119	0.8224	**0.0283**	0.7351	0.8646
	✓	✓	✓	**3.0398**	**0.8235**	0.0366	**0.6985**	**0.8705**
Mean	×	×	×	3.3966	0.8006	0.0468	0.7472	0.8816
	✓	×	×	3.1083	0.8274	0.0465	0.7004	0.8992
	✓	✓	×	3.1362	0.8508	0.0373	0.7167	0.9088
	✓	✓	✓	**2.9168**	**0.8570**	**0.0322**	**0.6416**	**0.9141**

feature distribution of the predicted time coarse-resolution image, thereby improving the spatiotemporal consistency of the random time fine-resolution image.

By averaging the metrics across several dates, we can see that with the addition of each module, the metrics gradually reach optimal values. This suggests that each module plays a crucial role in image transformation, feature extraction, feature fusion, and reconstruction. The more modules we add to the model, the better the overall performance of the network, ultimately leading to optimal fusion results.

5) Time Sensitivity Evaluation: In order to make fusion models more applicable in practical scenarios, reducing the sensitivity of spatiotemporal fusion models to the temporal information of auxiliary images has become a solution. Existing spatiotemporal fusion models typically require auxiliary images taken close to the prediction time, making them more reliant and sensitive to the temporal information of these images. The following experiments aim to explore the impact of auxiliary images with different temporal information on fusion models by predicting the same target date but using auxiliary images with varying time information. This is done to determine the degree and sensitivity of the model's reliance on temporal information. Table IV presents the quantitative evaluation results of the LGC dataset's test data on April 3rd using auxiliary images with different temporal information on January 29th, February 14th, and March 2nd. Table V shows the evaluation results of the CIA dataset's test data on May 4th using auxiliary images with different temporal information on April 11th, April 18th, and April 27th. From the metrics observed in Tables IV and V, it is evident that the two-input models, HPLTS-GAN and GAN-STFM, perform excellently regardless of the temporal information of the auxiliary images, with minimal differences in evaluation results across different auxiliary images. In contrast, there are significant performance differences among the other five models.

TABLE IV
QUANTITATIVE ANALYSIS OF THE LGC DATASET'S TEST DATA ON APRIL THIRD USING AUXILIARY IMAGES WITH DIFFERENT TEMPORAL INFORMATION

Model	Time Information	SAM	SSIM	RMSE	ERGAS	CC
STARFM	Jan.29th	4.6311	0.5745	0.0432	1.0691	0.7491
	Feb.14th	4.1028	0.6603	0.0456	1.0390	0.7970
	Mar.2nd	3.8605	0.7084	0.0352	0.8743	0.8258
FSDAF	Jan.29th	4.9927	0.6428	0.0470	1.1047	0.7468
	Feb.14th	4.1962	0.7374	0.0413	1.0504	0.8002
	Mar.2nd	3.8031	0.7907	0.0340	0.8633	0.8330
EDCSTFN	Jan.29th	3.7100	0.6434	0.0381	0.9660	0.7558
	Feb.14th	3.2366	0.7237	0.0349	0.9264	0.7902
	Mar.2nd	3.2346	0.7676	0.0326	0.8642	0.8011
GAN-STFM	Jan.29th	3.5932	0.6826	0.0336	0.8824	0.8112
	Feb.14th	3.3271	0.7410	0.0330	0.8193	0.8422
	Mar.2nd	3.1996	0.7753	0.0294	0.7871	0.8576
MLFF-GAN	Jan.29th	3.7498	0.6221	0.0477	0.9243	0.7729
	Feb.14th	3.4464	0.7043	0.0400	0.8493	0.8093
	Mar.2nd	3.2898	0.7745	0.0493	0.8128	0.8424
AMNET	Jan.29th	4.7556	0.5541	0.0434	1.1684	0.6774
	Feb.14th	3.8363	0.7015	0.0407	1.1234	0.7593
	Mar.2nd	3.3370	0.7695	0.0344	0.9338	0.8115
Ours	Jan.29th	3.5293	0.7381	0.0387	0.8121	0.8209
	Feb.14th	3.2356	0.7969	0.0393	0.7436	0.8541
	Mar.2nd	3.0398	0.8235	0.0366	0.6985	0.8705
Reference		0	1	0	0	1

As shown in Fig. 12, the performance of different models on the LGC and CIA datasets under different temporal information of auxiliary images is visually represented. The top and

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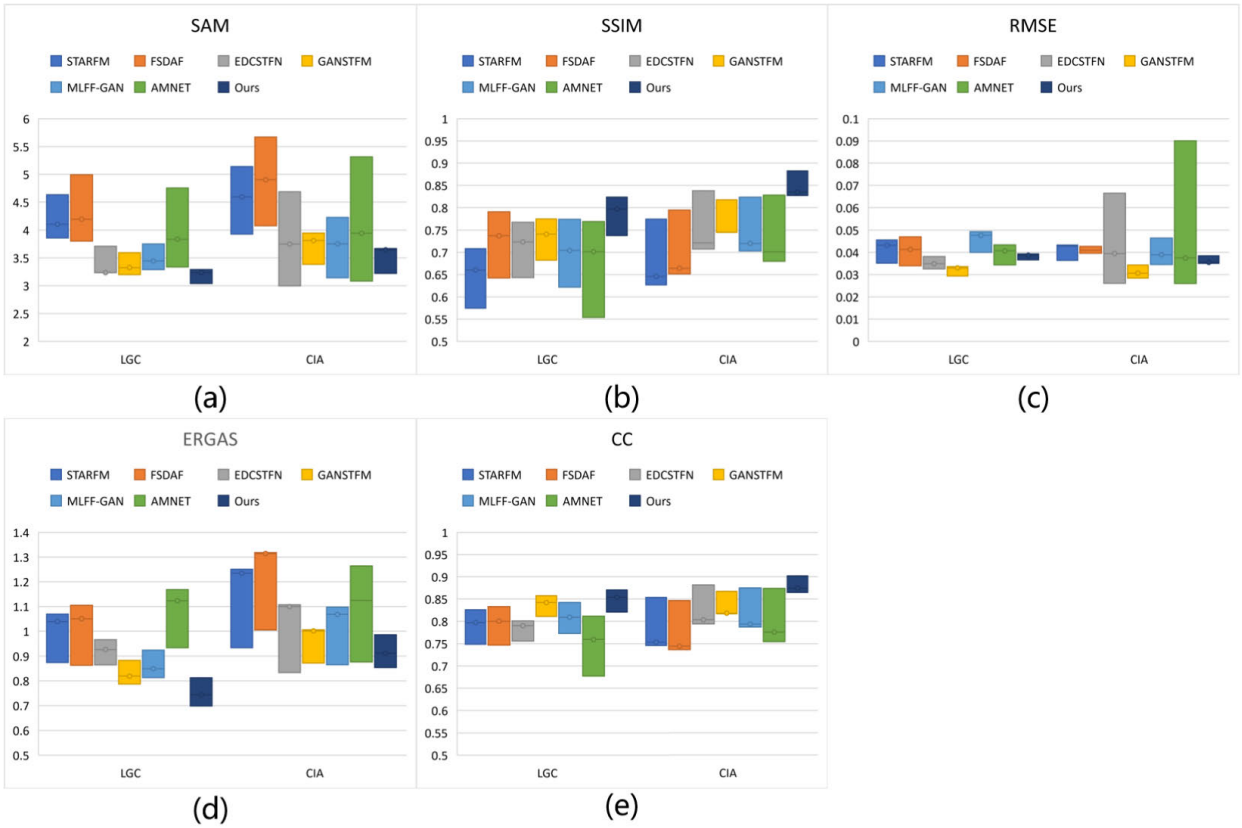


Fig. 12. Quantitative evaluation of temporal sensitivity of different models on LGC and CIA datasets, including (a) SAM index, (b) SSIM index, (c) RMSE index, (d) ERGAS index, and (e) CC index.

TABLE V
QUANTITATIVE ANALYSIS OF THE CIA DATASET'S TEST DATA ON MAY FOURTH USING AUXILIARY IMAGES WITH DIFFERENT TEMPORAL INFORMATION

Model	Time Information	SAM	SSIM	RMSE	ERGAS	CC
STARFM	Apr.11th	5.1399	0.6272	0.0433	1.2342	0.7463
	Apr.18th	4.5934	0.6458	0.0365	1.2508	0.7534
	Apr.27th	3.9295	0.7746	0.0427	0.9341	0.8536
FSDAF	Apr.11th	5.6718	0.6518	0.0410	1.3182	0.7367
	Apr.18th	4.9031	0.6643	0.0396	1.3140	0.7445
	Apr.27th	4.0788	0.7949	0.0427	1.0051	0.8465
EDCSTFN	Apr.11th	4.6863	0.7077	0.0395	1.1073	0.7942
	Apr.18th	3.7502	0.7210	0.0665	1.1000	0.8036
	Apr.27th	2.9966	0.8382	0.0261	0.8337	0.8819
GAN-STFM	Apr.11th	3.9473	0.7456	0.0343	1.0011	0.8170
	Apr.18th	3.8125	0.7450	0.0307	1.0055	0.8182
	Apr.27th	3.3849	0.8177	0.0284	0.8719	0.8673
MLFF-GAN	Apr.11th	4.2288	0.7035	0.0390	1.0969	0.7872
	Apr.18th	3.7547	0.7202	0.0345	1.0683	0.7939
	Apr.27th	3.1438	0.8235	0.0464	0.8648	0.8752
AMNET	Apr.11th	5.3132	0.6801	0.0375	1.2640	0.7547
	Apr.18th	3.9444	0.7013	0.0900	1.1246	0.7758
	Apr.27th	3.0836	0.8283	0.0260	0.8764	0.8740
Ours	Apr.11th	3.6699	0.8349	0.0351	0.9112	0.8745
	Apr.18th	3.6455	0.8276	0.0385	0.9855	0.8651
	Apr.27th	3.2238	0.8826	0.0355	0.8539	0.9023
Reference		0	1	0	0	1

bottom of each bar in the bar graph represent the maximum and minimum values of a certain model on a particular metric. The height of the bar (difference between the top and bottom) intuitively indicates the impact of auxiliary images with different temporal information on the performance of different models. We can observe that the heights of the bar graphs for the two-input models GAN-STFM and HPLTS-GAN are relatively small on both the LGC and CIA datasets. This suggests that the prediction accuracy of these models is less affected by the temporal information of auxiliary images. In contrast, the bar graphs for the other three-input models fluctuate in different sizes across different datasets and metrics, indicating that the prediction accuracy of the three-input models is more sensitive to temporal information. This also demonstrates that the two-input models GAN-STFM and HPLTS-GAN have lower dependency and sensitivity to temporal information. Compared to traditional methods, which strictly adhere to the temporal information of auxiliary images, two-input models have an advantage in terms of temporal sensitivity.

6) Parameters and Time Cost Evaluation: We implement the aforementioned deep learning-based spatiotemporal fusion models using the PyTorch framework and collect the parameter sizes and inference times of different networks. The inference time of HPLTS-GAN remains relatively optimal compared to other models, as shown in Table VI.

D. Discussion

The proposed method in this article demonstrates excellent fusion performance and a good visual representation of the experimental data and results from both the LGC and CIA

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TABLE VI
PARAMETERS AND TIME COST OF FIVE DEEP LEARNING METHODS

Model	Parameters	Time(s)
EDCSTFN	2.8×10^5	5.4
GANSTFM	5.8×10^5(Generator) 3.6×10^6(Discriminator)	10.7
MLFF-GAN	5.8×10^6(Generator) 3.6×10^6(Discriminator)	10.7
AMNET	2.8×10^5	5.4
Ours	5.8×10^7(Generator) 3.6×10^6(Discriminator)	10.7

datasets. The geological environments in the New South Wales region of Australia and the Lower Gwydir region of Gwydir significantly differ, yet our method performs well in both regions, indicating its robustness. This superiority is likely due to our model, HPLTS-GAN, being designed based on deep neural networks, with the inclusion of an ASDT module and an encoder–decoder structure. These components, respectively, address the issues of missing temporal information and significant resolution disparities between coarse and fine-resolution images in time-insensitive spatiotemporal fusion tasks. Thus, the HPLTS-GAN model can achieve better fusion results while mitigating dependency on temporal information.

Despite the improvements achieved by our method, there are still areas for further enhancement. First, in our method, extracting the relationship between the coarse-resolution image at the prediction time and the fine-resolution images at random times is crucial for improving spatiotemporal consistency in the fine-resolution images. Future work will focus on better utilizing the coarse-resolution image at the prediction time and extracting the relationship between the two to reduce the model's sensitivity to temporal information. Second, while our method employs MLFE, future consideration will be given to incorporating attention mechanisms at this stage to enhance focus on key information, thereby improving image quality and performance. In addition, while the design of the loss function in our method yields good results for the final fusion image, recent research CPW-STFN proposes a composite loss function incorporating wavelet loss for remote sensing image spatiotemporal fusion networks, which preserves spatial details effectively [32]. Future research directions will involve further attention to loss function design and integration of the wavelet loss composite loss function to further enhance the quality and performance of fusion images.

V. CONCLUSION

In this article, we propose a high-precision spatiotemporal fusion method that predicts fine-resolution images from randomly sampled times and coarse-resolution images predicted for a target time. The HPLTS-GAN model addresses the tradeoff between reducing dependence on time information and maintaining prediction accuracy. It improves the spatiotemporal consistency of fine-resolution images from random times, enhancing the overall performance of the model in time-insensitive tasks. The proposed model incorporates a U-Net-based encoder–decoder structure, enabling MLFE for different scale features. In the image reconstruction stage, an attention mechanism is used to selectively focus on the weights of shallow and deep feature information, allowing the network to preserve high-level semantic information while improving the recognition of details and overall model performance.

HPLTS-GAN is compared with existing methods such as STARFM, FSDAF, EDCSTFN, GAN-STFM, MLFF-GAN, and AMNET through metrics, visual comparisons, and residual map analysis. The results confirm that HPLTS-GAN achieves high prediction accuracy in the two-input model and competes well with three-input models. In addition, ablation experiments are conducted to evaluate the effectiveness of the multilevel feature, CSAFF, and ASDT modules. The multilevel feature module effectively reduces information loss, the CSAFF module adjusts the weights of different-level features, enhancing the model's perception of detailed information, and the ASDT module improves the spatiotemporal consistency of fine images from random times, resulting in better performance in abrupt change regions and overall spectral and texture features. In summary, comparisons and ablation experiments both confirm that HPLTS-GAN excels in reducing time dependence while maintaining high prediction accuracy, giving it an advantage over the other six methods in practical applications.

Despite the achievements, there is still considerable room for improvement in spatiotemporal fusion methods based on time-insensitive remote sensing. Future work in this article will continue to focus on understanding the relationship between the coarse-resolution image at the prediction time and the fine-resolution image at random time, further reducing the model's sensitivity to temporal information. In addition, attention will be given to incorporating attention mechanisms in the feature extraction stage to enhance feature extraction and aid in generating higher-precision images. Finally, attention will be paid to exploring new loss functions to improve the quality of fused images.

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